Risk Assessment and Mitigation

Cohort 3
Group 11
Team Aubergine
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Risk Register

During the risk analysis, our team identified any potential risks or threats to our development process in terms of our project or our product and any ways to mitigate the effects of these risks. Our risk engineering approach was done over the course of our project, adding in risks as we saw fit. We made sure the risks were genuine risks that had to be mitigated in likelihood, not by solutions as if a risk can be solved it is no longer a risk. We identified the risks by identifying ongoing problems that could not be easily solved.

Our risk table outlines the processes risk analysis taken to assign it a category, explain the risk, give it a rating for severity and likelihood, and outline the mitigation strategies and ownership of each specific risk. To categorise the risks we considered the potential effect of the risk as it would give us a clear indication at what each risk focused on and helped with the risk strategy process by allowing us to consider what effect the risk could have if not mitigated appropriately. This means looking at whether the risk would affect the state of the project, future development plans or overall team work. In addition, we used risk likelihood and severity as metrics to measure the potential threat of a risk to the development process. This was done because it shows a quick view of how severe a risk is and therefore how much focus should be put on the mitigation of said risk. In addition it would allow us to evenly distribute ownership for risks and avoid having too many high severity risks owned by 1 team member.

We focused on describing and refining each risk. For every risk we:

1. Discussed its category — project, product and project, technology, or business.
    a. Project meaning risks that will impact the process of production
    b. Product meaning risks that will impact the quality and completeness of the end product
    c. Product and project meaning risks arising from changes and delays to requirements and specifications
    d. Business meaning risks from procuring/ developing the product.
2. Set a likelihood rating using the L/M/H scale: H (High, probably will happen), M (Medium, not unlikely but may not happen), L (Low, probably won't happen)
3. Set a severity rating: S (Significant), T (Tolerable), I (Insignificant);
4. Defined mitigation measures.

Each risk was assigned to an owner responsible for that risk. We also agreed on ownership distribution so that no single person manages multiple high-severity risks.

ID	Type	Description	Likehood	Severity	Mitigation	Owner
R1	Project	Build up of code changes making merges difficult	M	I	Make small commits often to avoid large merges	Harrisio n

R2	Project	The lack of online communication in the team	M	S	WhatsApp chats where we can communicate and share feedback.	Harry
R3	Project	Burnout during the project	M	T	Assigned roles and deadlines for each team member to complete their part of the work.	Harry
R4	Product	Scope creep in implementation stage	H	S	Regular discussions within implementation about necessary features and requirements	Harry
R5	Product and project	Requirement change from the user	H	S	Maintain constant communication with client and agile development methodology	Sarvesh
R6	Project	Team member falls ill	H	I	Tasks delegated early on and split up to ensure work can be done from home if needed. Using whatsapp to inform group members	Harry
R7	Product	Key functionality not possible with chosen libraries	L	T	Find work around to either fulfil the requirement in a different way or use a different library	Arnav

R8	Product and project	Delay in key requirement	H	T	Weekly updates of the plan to take delays into consideration	Arnav
R9	Project	Software tools not working	H	T	Use different tools if possible, else work with team to get working	Sarvesh
R10	Product and project	Key misinterpretation of requirements	M	S	Regular client communication and agile methodology. Review requirements periodically to understand better	Arnav
R11	Project	Project runs over testing time	H	I	Test code during development to maintain quality without dedicated testing	Sarvesh
R12	Project	Breakdown of team communication	M	S	Communicate with all team members with weekly meetings	Harry
R13	Project	Team disagreements in task execution may cause delays or inconsistent results	M	S	Clarify roles in advance and hold regular team meetings to resolve conflicts.	Callum

R14	Project	Group low cohesion with communication between technical and non-technical team	M	S	Communicate required jobs via whatsapp and document jobs needed to do in a jobs list	Harry
R15	Business	Breakdown in continuous Integration with new project	M	S	Document tools used and refer back to previous documentation frequently	Harry